LAB 6

TITLE: DYNAMIC ROUTING CONFIGURATION: RIP, EIGRP, OSPF, BGP

Objectives

- To understand the working principles of RIP, EIGRP, OSPF, and BGP routing protocols.
- To configure dynamic routing protocols on routers using Cisco Packet Tracer.
- To enable automatic route discovery between different networks.
- To establish communication between networks using dynamic routing configurations.

Theory

1. RIP (Routing Information Protocol)

Operation: RIP employs a distance vector routing algorithm using hop count as its metric, with maximum limit of 15 hops. The protocol broadcasts complete routing tables to directly connected neighbors every 30 sec and relies on the Bellman Ford algorithm to calculate optimal routes.

Application: It is ideal for small scale networks with straightforward topologies. Its frequently deployed in small office/home settings and educational laboratories where ease of configuration outweighs the need for advanced features.

Limitation: Scalability constraint: 15-hop maximum (16 hops indicates an unreachable network). Slow convergence: Takes considerable time to adapt to topology changes

2. EIGRP (Enhanced Interior Gateway Routing Protocol)

Operation:EIGRP uses the DUAL algorithm for loop-free routing. It calculates metrics based on bandwidth, delay, load, and reliability. Sends partial, triggered updates only during topology changes and maintains neighbor relationships via hello packets.

Application Suitable for large enterprise networks requiring fast convergence and scalability. Supports hierarchical designs and multi-vendor environments through route redistribution.

Limitations Cisco-proprietary protocol with limited interoperability, complex configuration, higher CPU and memory requirements, and challenging troubleshooting in large deployments.

3. OSPF (Open Shortest Path First)

Operation: OSPF employs a link state routing mechanism, creating a comprehensive topology map of the entire network. It applies Dijkstra’s algorithm to compute optimal routes and divides networks into distinct areas, with Area 0 functioning as the central backbone. Updates are shared through link state advertisements while neighbor relationships are actively monitored.

Application: Perfect for expansive enterprise environments where rapid adaption to network changes and extensive scalability are priorities. Excels in hierarchical architectures with area segmentation and performs well in topologies featuring multiple alternative paths.

Limitation: Demanding setup and diagnostic procedures, substantial processing power and memory overhead during route calculations necessitates strategic area architecture and design, challenging mastery curve for IT professionals.

4. BGP (Border Gateway Protocol)

Operation: BGP is a path vector exterior gateway protocol designed to exchange routing information across autonomous systems AS. It establishes reliable connections via TCP port 179 and makes intelligent routing decisions based on administrative policies, As path attributes, and business rules rather than simple metrics. The protocol maintains persistent peering sessions between BGP routers.

Application: Critical for global internet infrastructure and inter domain routing. Enables organizations to connect with Internet Service Providers, implement multi homed network configurations perform advanced traffic engineering, and enforce policy driven routing decisions between different administrative domains or autonomous systems.

Limitation: Extremely complex setup demanding extensive protocol knowledge and expertise, slower convergence times relative to interior gateway protocols, requires substantial memory to store complete internet routing tables (800,000+ prefixes), highly sensitive to misconfigurations that can trigger widespread internet disruptions,

Network Topology

A network topology was created using two routers: Router1Binay(1841) and Router2Binay (1841), interconnected via a serial/WAN link using their serial interfaces. The topology consists of two separate LANs connected through these routers.

Switch0 is connected to Router1Binay and provides connectivity to two end devices: PC0 and PC1 forming network 192.168.1.0/24. Switch1 is connected to Router2Binay and provides connectivity to two end devices: PC2 and PC3 forming network 192.168.2.0/24.

The serial connection between Router1Binay and Router2Binay enables inter-network communication, allowing devices in the 192.168.1.0/24 network to communicate with devices in the 192.168.2.0/24 network through dynamic routing protocols. Green indicators on the connections represent active and functioning network links in the Cisco Packet Tracer simulation environment

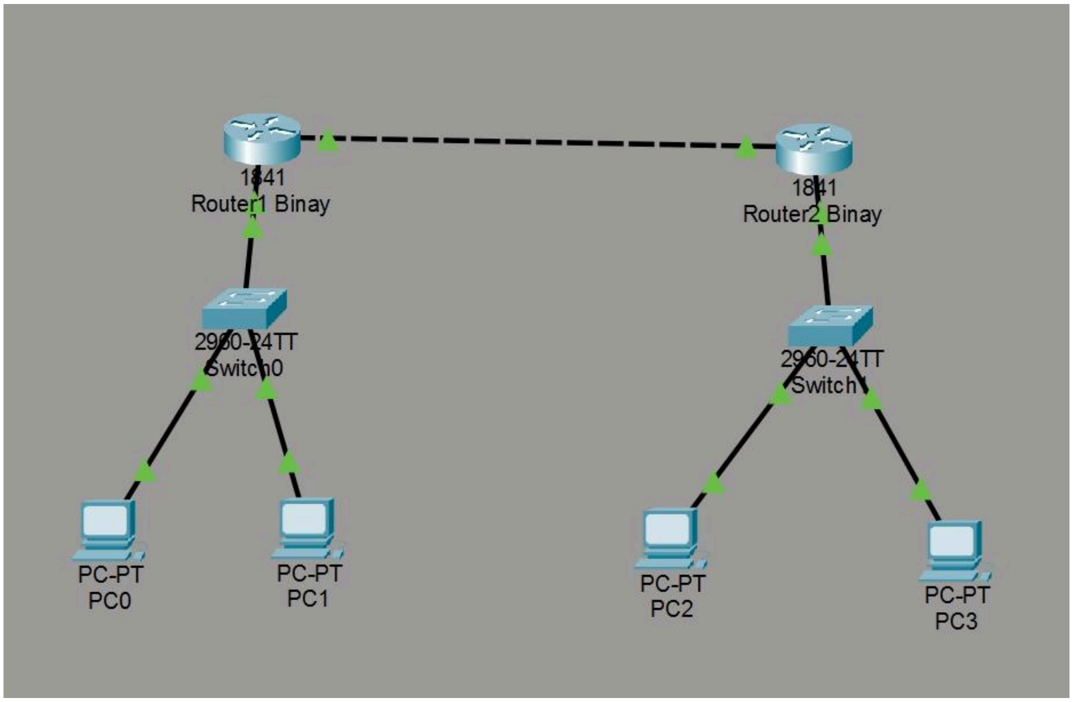

Fig: Network Topology

Configuration Table:

Network devices were configured with IP addresses and subnet masks as detailed in the table below:

IP Configuration Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router1Binay	Gig0/1	192.168.1.1	255.255.255.0	N/A
Router1Binay	Gig0/0	10.0.0.1	255.0.0.0	N/A
Router2Binay	Gig0/1	192.168.2.1	255.255.255.0	N/A
Router2Binay	Gig0/0	10.0.0.2	255.0.0.0	N/A
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1

1. RIP Configuration Commands

Router2 (Left sided one in topology):

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

Note: If you just see “Router >” instead of “Router(config)#” (as in the first line), then execute the following commands also.

```
Router>enable
Router# configure terminal
Router(config)#
```

Router1 (Right sided one in topology):

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.2.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

2. EIGRP Configuration

Commands Router2 (Left sided one in

topology):

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.1.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

Router1 (Right sided one in topology):

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.2.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

3. OSPF Configuration Commands

Router2 (Left sided one in topology):

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

Router1 (Right sided one in topology):

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

4. BGP Configuration Commands

Router2 (Left sided one in topology) (AS 65001):

```
Router(config)# router bgp 65001
Router(config-router)# neighbor 10.0.0.2 remote-as 65002
Router(config-router)# network 192.168.1.0 mask 255.255.255.0
```

Router1 (Right sided one in topology) (AS 65002):

```
Router(config)# router bgp 65002
Router(config-router)# neighbor 10.0.0.1 remote-as 65001
Router(config-router)# network 192.168.2.0 mask 255.255.255.0
```

Result

Dynamic routing protocols were successfully deployed and validated across the network infrastructure. Connectivity tests were conducted by transmitting ping packets from PC2 (192.168.2.2) to PC0 (192.168.1.2), spanning both network segments. Each protocol—RIP, EIGRP, OSPF, and BGP—effectively learned routes automatically and maintained accurate routing tables. All ping requests achieved 100% success rate with zero packet loss, confirming seamless communication between networks and verifying proper protocol configuration and functionality.

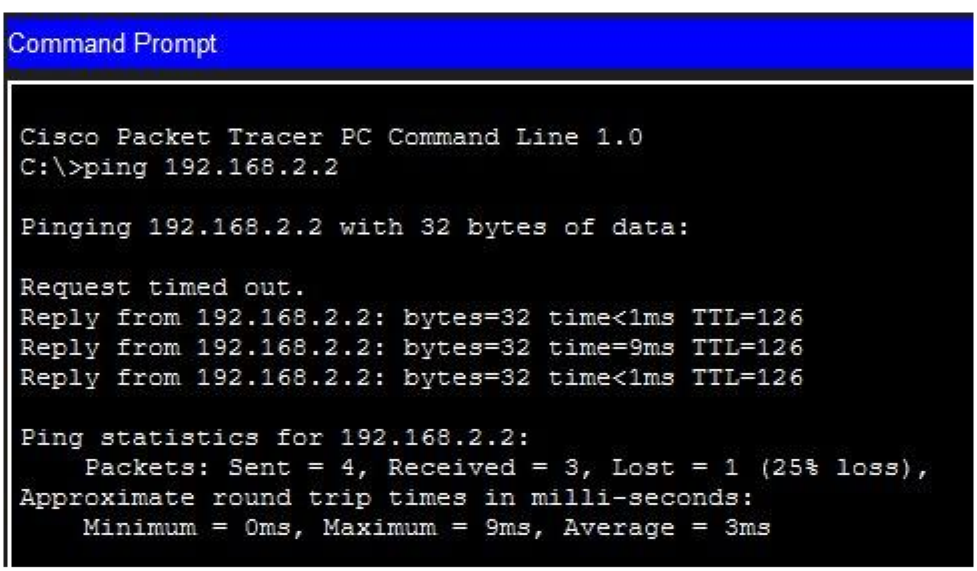

Fig: Ping (RIP configuration)

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Fig: Ping (EIGRP)

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=9ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 3ms
```

Fig: Ping (OSPF)

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Fig: Ping (BGP)

Discussion

The experiment successfully demonstrated four dynamic routing protocols in a simulated Cisco environment. RIP offered simple configuration but slower convergence with periodic updates. EIGRP provided faster convergence through event-triggered updates and reduced bandwidth usage. OSPF required hierarchical area configuration but delivered vendor-independent standardization. BGP introduced the highest complexity with autonomous system requirements and policy-based routing. Successful ping tests between PC0 and PC2 confirmed automatic route discovery and seamless inter-network communication across all protocols.

Conclusion

The laboratory achieved its objective of implementing and comparing dynamic routing protocols (RIP, EIGRP, OSPF, and BGP) on Cisco routers. All protocols were successfully configured and verified through connectivity tests, demonstrating automatic route learning and dynamic routing table maintenance without manual static routes.